

Sprint Action Plan: Person 1: Theory Lead

Mathematical Formulation & Adaptive Kernel Logic • Adaptive Attention Computation Sprint

Assigned Role: Person 1: Theory Lead	Estimated Workload: 9.0 Hours	Sprint Timeline: 0.0h – 9.5h
Core Discipline: Mathematical Modeling / Kernel Theory	Primary Handoff To: Person 4 (Model Patch) & Person 6 (Deck)	Target Hardware: Dual T4 (16GB)

1. Role Mission & Strategic Objective

Formulate the analytical error certificate linking positive-kernel approximation error to attention output accuracy. Design the dynamic rank selection rule $r^*(X, \delta)$ to balance computational efficiency against key-value retrieval quality. Verify numerical stability safeguards and provide mathematical rigor for all project claims.

2. Input Dependencies & Final Deliverables

Type	Item / Specification	Source / Destination	Deadline / Gate
INPUT	Pre-registered hypothesis & variables	Team 3 Revised Proposal	Hour 0.0
INPUT	Empirical retrieval collapse numbers	Person 3 (Retrieval Lead)	Hour 6.0
OUTPUT	Mathematical error bound specification	Team Repository (theory_spec.md)	Hour 1.0
OUTPUT	Adaptive-rank kernel logic (adaptive_kernel.py)	Person 4 & Person 2	Hour 7.5
OUTPUT	Theory slide narratives (Slides 2, 5, 6)	Person 6 (Synthesis Lead)	Hour 8.0

3. Technical Specifications & Contract Signature

Key Code / Mathematical Contract for Person 1: Theory Lead:

- Error Certificate Condition: $\epsilon_r(X) \leq \delta * \min_i(Z_i) / [N * (2*V_{max} + \delta)]$
- Dynamic Rank Rule: $r^*(X, \delta) = \min\{ r \in \{64, 256, 1024\} : B(X, r) \leq \delta \}$
- Analytical Toy Check: $QK^T = [[1,2,3],[2,1,3],[1,1,2]]$; Linear output = [4.8, 5.2] vs Softmax = [4.28, 5.72]
- Kernel I/O Contract: `def adaptive_rank_attention(Q, K, V, delta=0.05, r_candidates=(64, 256, 1024), is_causal=False)`

4. Hour-by-Hour Master Action Checklist — Person 1: Theory Lead

Window	Specific Actions & Step-by-Step Instructions	Deliverable / Evidence	Status
0.0 – 1.0 h	Lock tensor I/O contract; publish mock kernel stub; finalize mathematical error bound formulation.	Frozen function signature + MockKernel	[] Done
1.0 – 3.0 h	Derive output error bounds; verify 3-token analytical toy example; compute exact relative Frobenius discrepancy.	Mathematical verification report	[] Done
3.0 – 5.5 h	Implement adaptive rank selection algorithm based on query-key entropy and logit temperature.	adaptive_kernel.py (Rank selection logic)	[] Done
5.5 – 7.5 h	Unit-test kernel for numerical stability, FP32 accumulator safeguards, and denominator epsilon clamping.	Unit test suite passing 100%	[] Done
7.5 – 8.5 h	Deliver production kernel to Person 4; review mathematical writeup with Person 6.	Kernel handoff + theory appendix	[] Done
8.5 – 9.5 h	Rehearse mathematical Q&A; defense for departmental presentation; verify final report equations.	Presentation sign-off	[] Done

5. Operational Guardrails & Failure Protocols

Constraint / Risk	Enforcement Rule & Exact Action
FP16 Accumulator Overflow	Enforce FP32 accumulators for all summations across N to prevent Inf/NaN in long context.
Zero Denominator	Clamp denominator with $\phi(Q) @ \phi(K)^T 1 + 1e-6$ to guarantee mathematical stability.
Associativity Limits	Never claim $(QK^T)V = Q(K^TV)$ holds for softmax; clearly explain the kernel substitution.

6. Definition of Done (DoD) & Sign-Off Checklist

[]	Analytical 3-token toy calculation numerically verified ([4.8, 5.2]).
[]	Error bound certificate formula validated against empirical sweeps.
[]	adaptive_kernel.py adheres strictly to the frozen function signature.
[]	Slide 2, 5, and 6 narratives and LaTeX equations reviewed and signed off.